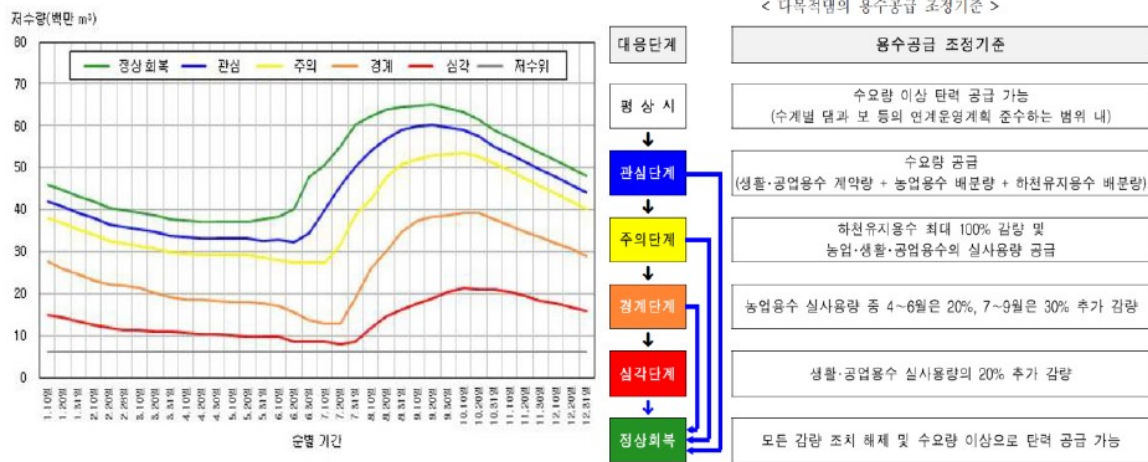


Click on - / - - - - .

Your data should be the DAILY observation data from 1998-01-01 through 2017-12-31.

- D. Organize the data so that they are in the correct order. Click on (Cell B3). Then, click on - - . Check if your data is in the correct order from 1998-01-01 through 2017-12-31.
 - E. Among the data, this time you will only use the following data:
(1) (m3), (2) (m3/s), and (3) (m3/s) along with the date.
Copy only the columns in B, E, H, and K into another file named
“BR_Simulation.xlsx” under the Sheet named “BR_Data”.
 - F. Before moving on to the next step, transform the data in Column B from text into numbers.
5. You will create a daily model of Boryeong Dam in Microsoft Excel by following the steps below:
- A. First, extract the inflow data between 2016-10-01 through 2017-09-30. Save the data into another sheet named “2017_WY_Sim.” Transform the scale of the inflow into MCM (Million cubic meters) in Column H.
(Please note the scale of the storage is different from the inflow and the outflow in BR_Data Sheet)
 - B. Transform the monthly demand data of Boryeong Dam into MCM. Roughly, the aggregate demand of Boryeong Dam should be 100 MCM annually.
 - C. Using the “vlookup” function in Excel, create a column with daily demand corresponding to each month. Check if your demand changes in June.
 - D. Assume that no matter what, release is equal to demand () and run a simulation of Boryeong Reservoir according to the flow diagram. What are the problems you observe? When is the first time your model observes negative storage?
 - E. In order to prevent the negative storage problem, modify your model so you’re your model satisfy the feasibility constraints ($S_{min} \leq S_t \leq S_{max}$)
(Hint: use Min/Max functions in Excel)
 - F. Re-calculate actual release that satisfy the feasibility constraints and the continuity equations. Check if your model meets the annual mass balance equation (Total Inflow + Starting Storage = Total Actual Outflow + Final Storage). What is the sum of the left-hand side equation? Right-hand side of the equation? (They should be very much similar to each other)
 - G. Finally, refer to the following < > that are being applied to all reservoirs in Korea. The main concept of < > (hedging) is in reducing the release prior to low storage. How do you think the implementation of hedging rules during operation affect the performance of the reservoir? Why do you think like that?



〈그림 2〉 보령댐 용수공급조정기준

- H. Simply, in order to observe the effect of hedging, create a model that reduces release by 50% when the storage falls below 25 MCM. Draw a graph that compares the release and storage of Boryeong Dam with and without hedging. Do your assumptions made in G follow the conclusions made from the simulations in H?
- I. What are the key assumptions you made in building the model? List at least three key assumptions inside the model.
- J. How will you enhance the model's practicality?
()

457.557 Water Resources Systems Engineering, Spring 2022

Assignment #2

Due before the lecture of Apr-19, 2022

1. Solve and attach your solving procedures on the following problems on the textbook:

- A. 4.1 (a), (b), (c), & (d)- Engineering economics
- B. 4.9 (a) & (b)- Lagrange multiplier
- C. 4.15- Dynamic programming (deterministic)
- D. 4.18- Dynamic programming

2. Solve the following LP Problem in Excel and in R:

$$\text{Maximize } Z = 10x_1 + 8x_2 + 9x_3$$

$$\text{Subject to: } 2x_1 + 3x_2 + x_3 \leq 1000$$

$$5x_1 + 6x_2 + 6x_3 \leq 2400$$

$$x_1 + x_2 + x_3 \leq 700$$

$$x_1 \geq 0, \quad x_2 \geq 160, \quad x_3 \geq 0$$

- A. Attach your R code.
- B. Compare your results obtained from Excel and R. Are they identical?

3. Solve the following DP problem in R:

The inflows to a reservoir having a total capacity of four units of water are 1, 2, 1, and 2 units, respectively, for the four seasons in a year. For convenience, the amount of water is counted only by integer units of water. Thereby, the releases from the reservoir for water supply to a city and farms are sold for \$2000 for the first unit of release, \$1500 for the second unit, \$1000 for the third unit, and \$500 for the fourth unit. When the reservoir is full and spills one unit of water, a minor flood with result in a damage of \$1500. When the spill reaches 2 units, a major flood will result in a damage of \$4000. Determine the operation policy for maximum annual returns by dynamic programming using a backward algorithm, considering any amount of storage at the end of the year.

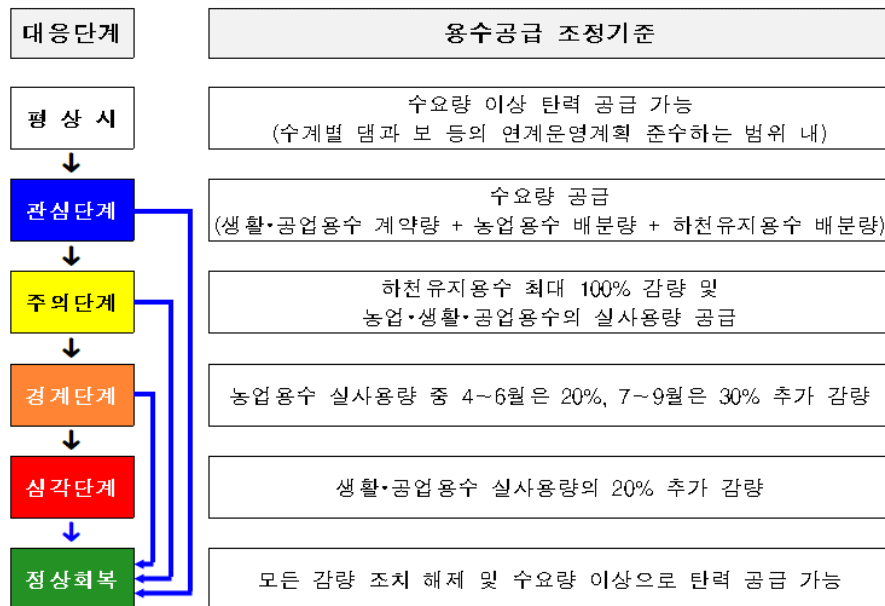
- A. First, set the benefit function of your problem. How would you set your benefit function?
- B. What is the governing equation of your problem?

- C. What is the DP backward recursive equation for this example? Your equations should be different for seasons 1-3, and season 4.
- D. Solve your problem in R. You may solve your problem first using Excel first.
- E. Assume that the starting storage (S0, before Stage 1) is 4. What are the optimal release decisions during the four seasons? There should be multiple optimal release decisions. Name all and compute each path's annual return values.
4. In R, replicate the reservoir model on Boryeong Dam by you built on Assignment #1 with actual zone-based hedging rules provided below:

1.10	43.9	39.9	30.1	16
1.20	42.4	38.3	28.4	15.1
1.31	40.8	37.2	26.8	14.1
2.10	39.5	35.5	25.3	13.2
2.20	38	34.4	24.7	12.4
2.28	37	32.9	23.8	11.8
3.10	36	31.9	22.9	11.7
3.20	35	31.5	22.3	11.5
3.31	34.5	30.5	21.7	11.3
4.10	34.4	30.2	21.2	11.3
4.20	34.2	30	21.1	11.1
4.30	34.1	30.3	20.4	10.9
5.10	34.1	30.3	19.9	10.6
5.20	34.3	30.4	18.9	10.3
5.31	34.2	30.4	18.4	10.3
6.10	34.5	30.3	17.8	10.2
6.20	35.3	29.6	16.4	9.2
6.30	38.2	29.5	15.1	9
7.10	43.3	30.5	14.7	9
7.20	48	35.8	17.4	8.6
7.31	52.7	40.8	23.2	9.6
8.10	56.8	45.1	29.6	12.9
8.20	59.7	50.3	33.6	15.2
8.31	61.6	53.4	38.7	17.1
9.10	62.6	54.6	40.8	18.4
9.20	62.6	55.2	41.2	19.9
9.30	61.9	55.5	41.9	21.2
10.10	60.7	55.6	42.6	22.3
10.20	59.1	54.5	42.6	22.3
10.31	57	53	41.3	22.1
11.10	55.1	51	39.5	21.4
11.20	53.2	49.1	37.7	20.4
11.30	51.3	47.2	36.3	19.4
12.10	49.3	45.3	34.9	18.5
12.20	47.6	43.5	33.4	17.7
12.31	45.7	41.6	31.3	16.8

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< 다목적댐의 용수공급 조정기준 >



- First, interpolate daily values with the provided values. You should use linear interpolation method in R. You do not have to include your interpolation results.
- Assume that at the _____ stage, water supply for M&I() water is 3.0m³/s instead of 2.9m³/s. Conduct a reservoir simulation using the daily _____ you interpolated in A. Your model should consider the maximum and minimum constraints.
- Check if your model meets the annual mass balance equation (Total Inflow + Starting Storage = Total Actual Outflow + Final Storage). What is the sum of the left-hand side equation? Right-hand side of the equation? (They should be very much similar to each other)
- Finally, compare your results with results from no hedging. What is the effect of hedging? Qualitatively? Quantitatively?